Hingna Road, Wanadongri, Nagpur.

Odd Term—2026-27
Semester – V
Subject Code : 23IT1501
Time :1½ Hours

Mid Semester Exam - II
Date: - 28/10/2026
Subject : Theory of Computation
Max. Marks: 30

Note:-
1) Attempt all questions.
2) Figures in bracket to the right indicate the marks for questions.
3) Marks will be allotted as zero if wrong question number is mentioned.

| | | Max Marks | CO Level |
|---|---|---|---|
| **Que 1** | **Solve any two of following** | | |
| | (a) Find the transition rules of PDA for accepting a language $L=\{w \in \{a,b\}^* \mid w \text{ is of the } a^n b^n \text{ with } n \geq 1\}$ through both empty stack and final state and Demonstrate the stack operation for the string aaabbb | [5] | CO2 (Level 6) |
| | (b) Design a PDA for accepting a language $\{a^n b^{2n} \mid n>=1\}$ . Simulate this PDA for the input string "aaabbbbbb". | [5] | CO2 (Level 6) |
| | (c) Construct a PDA equivalent to following CFG<br>$S \rightarrow BD \mid BC$<br>$D \rightarrow SC$<br>$C \rightarrow AA$<br>$B \rightarrow 0$<br>$A \rightarrow 1$ | [5] | CO2 (Level 6) |
| **Que 2** | **Solve any two of following** | **Max Marks** | **CO Level** |
| | (a) Define a Turing machine. Construct a TM that accept $L = \{wcw^R \mid w \in (0, 1)$ and c is ε or 0 or 1. Show that string 0110 is accepted by this TM with sequence of Instantaneous Description (ID) | [5] | CO3 (Level 6) |
| | (b) Construct a Turing machine that accepts the language:<br>$L(M) = \{ww \mid w \in \{a, b\}^*\}$<br>Give Suitable Example with of Instantaneous Description (ID) | [5] | CO3 (Level 6) |
| | (c) Design a Turing Machine that performs the subtraction of two unary integers. Given two strings on the Turing tape as input, the machine should halt in the accept state with the result that is the number of elements of string one minus the number of elements in string two. The machine needs only to consider cases where the result is zero or positive. Draw the transaction diagram and Diagram and transition table also provide all the rules. | [5] | CO3 (Level 6) |
| **Que 3** | **Solve any two of following** | **Max Marks** | **CO Level** |
| | (a) Find whether the lists M = (B, A, CA ,ABC) and N = (CA, AB, A, C) have a Post Correspondence Solution ? | [5] | CO4 (Level 5) |
| | (b) Compute A(1,1),A(2,1),A(1,2) and A(2,2) by using Ackermann function and justify is it the recursive enumerable function. | [5] | CO4 (Level 5) |
| | (c) Prove that addition and mn of two integer are primitive recursive also explain when a problem is said to be decidable or undecidable? Give an example of an undecidable problem | [5] | CO4 (Level 3) |

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Nagar Yuwak Shikshan Sannstha's
**Yeshwantrao Chavan College of Engineering**
**(An Autonomous Institution Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)** Hingna Road, Wanadongri, Nagpur.

ODD Term—2026-27
Semester – Vth
Subject Code : 23IT1506
Time :1½ Hours

Mid Semester Exam - II
Date:- 30/10/26
Subject: Operating Systems
Max. Marks: 30

**Instructions:**

1. Solve any two from Q1,Q2,Q3 each.
2. Give appropriate figure where ever necessary
3. Marks will be allotted as zero if wrong question number is mentioned.

| | | | |
|---|---|---|---|
| Q.1. | A) Compare sequential, direct, and indexed access methods. Provide examples of where each is used | [05] | CO3 (Level 2) |
| | B) Suppose that disk drive has 5000 cylinders from 0 to 4999.The drive is currently serving request at cylinder 143 and previous was at cylinder 125.The pending request in FIFO is 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from current position what is the total distance that disk arm moves to satisfy all the Pending request for each of following disk scheduling algorithms?<br>i) FCFS ii) SSTF iii) SCAN | [05] | CO3 (Level 3) |
| | C)Recommend a suitable disk-scheduling algorithm for a heavily loaded system and justify your choice. | [5] | CO3 (Level 2) |
| Q.2. | A)A memory system contains five free memory blocks of sizes 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB, respectively. Four processes of sizes 212 KB, 417 KB, 112 KB, and 426 KB arrive sequentially. Apply the First Fit and Best Fit and Worst memory allocation algorithms to allocate the given processes. Draw the memory allocation diagram for each algorithm and determine which technique successfully allocates the maximum number of processes. Justify your answer. | [05] | CO3 (Level 3) |
| | B)Explain the concept of paging in memory management. How does it eliminate external fragmentation? | [05] | CO3 (Level 2) |
| | C)Explain the concept of TLB hit and TLB miss. How do they affect system performance? | [05] | CO3 (Level 2) |
| Q.3. | A) Explain the concept of Virtual Memory. Discuss the advantages and disadvantages of demand paging in virtual memory management. | [05] | CO4 (Level 2) |
| | B)Consider the following page reference string 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 for a memory with 3 frames. How many page faults would occur for the following page replacement algorithms?<br>i) LRU ii) FIFO | [05] | CO4 (Level3) |
| | C)Explain the concept of Belady,s Anamoly in FIFO page replacement algorithm with proper illustration. | [05] | CO3 (Level2) |

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**Yeshwantrao Chavan College of Engineering, Nagpur**
**Department of Information Technology**
**ODD Term (Session 2026-27)**
**Mid Semester Examination-II**
**Subject: Machine Learning [23IT1504]**

**Time: 1.5 Hrs.** **Date:26.10.26** **Max. Marks: 30**

**Solve any two sub questions from each question**

| | | | | | |
|---|---|---|---|---|---|
| Qu 1 | a) | Cluster the following two-dimensional data into two clusters using k-means clustering algorithm: (182,72), (170,56), (168,60), (179,68), (182,72), (188,77). The underlined data points are the initial cluster centers. Use Manhattan distance. Perform two iterations only. | 05 | CO2 | L3 |
| | b) | What is divisive clustering. Perform divisive hierarchical clustering using the Minimum Spanning Tree (MST) approach for the following data points. Use Euclidean distance.<br>Data Point: A (1,1), B (2,2), C (5,2), D (6,3) | 05 | CO2 | L3 |
| | c) | What is a dendrogram? Perform agglomerative clustering with single linkage and Euclidean distance on the following 1-D dataset {7,10,20,28,35}. Also Construct the dendrogram. | 05 | CO2 | L3 |
| Qu 2 | a) | What is k-fold cross validation and why is it required?<br>Suppose a dataset contains 300 labeled samples, denoted by (X1, X2, ... X300). A machine learning engineer applies 3-fold cross-validation to evaluate a model with a fixed hyperparameter configuration.<br>i) How many times is the model trained and evaluated?<br>ii) How many samples are used for training and validation in each fold?<br>iii) How should the engineer evaluate the model's overall performance based on the cross-validation results? | 05 | CO3 | L4 |
| | b) | Two machine learning regression models, MR1 and MR2, have been developed to predict the length of hospital stay (in days) for patients based on their clinical characteristics. The actual length of hospital stay for five patients is: y_true = [5,7,9,4,8]. The predicted values obtained from the two models are:<br>y_pred_MR1 = [4.8,6.7,8.5,4.2,7.6]<br>y_pred_MR2 = [4.6,6.5,8.8,4.4,7.3]<br>Answer the following questions:<br>i) Calculate the Mean Squared Errors for regression model MR1 and MR2<br>ii) Compare the performance of MR1 and MR2 on the given test data. Which model performs better? Justify your answer. | 05 | CO3 | L4 |
| | c) | What is a confusion matrix? Consider the following confusion matrix of a ML classifier, and answer the following questions:<br>(i) What is the size of the dataset?<br>(ii) Is it a symmetric dataset? Justify.<br>(iii) Find the accuracy and F1 Score of this model.<br>Prediction: Cat, Dog; Actual: Cat (15, 35), Dog (40, 10) | 05 | CO3 | L4 |
| Qu 3 | a) | An MP neuron is configured to implement a 3-input NOR gate with weights w1 = w2 = w3 = - 1 and threshold = -1. Evaluate whether the given configuration correctly implements a 3-input NOR gate. Justify your answer using the truth table. If the configuration is incorrect, recommend suitable modifications. | 05 | CO4 | L5 |
| | b) | A CNN uses pooling layers to reduce the spatial dimensions of feature maps.<br>Consider the following feature map.<br>i) Apply Average Pooling and Max Pooling separately using a 2x2 window and a stride of 2.<br>ii) Evaluate which pooling technique is more appropriate for preserving the strongest feature activations. Justify your answer using the computed feature maps.<br>3 2 6 5<br>2 1 6 9<br>7 9 4 6<br>7 8 2 5 | 05 | CO4 | L5 |
| | c) | A neural network is designed to classify inputs into four mutually exclusive classes. For a given input, the final fully connected layer produces [1.1, 2.05, 1.8, 0], as scores, before Softmax activation.<br>i) Calculate the output probability vector obtained after applying the Softmax activation function.<br>ii) Identify the class with the highest predicted probability.<br>ii) Evaluate the suitability of Softmax and Sigmoid activation functions for the given four-class classification problem. Recommend the more appropriate activation function and justify your choice based on the nature of the classes and output probabilities. | 05 | CO4 | L5 |

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Hingna Road, Wanadongri, Nagpur.

ODD Term—2026-27
Semester – V
Subject Code: 23IT1502
Time: 1½ Hours

Re-Mid Semester Exam
Date: - 04/11/2026
Subject: Database and Information System
Max. Marks: 30

**Instructions:**
1. All questions are compulsory
2. Give appropriate figure where ever necessary
3. Marks will be allotted as zero if wrong question number is mentioned.

| | | | |
|---|---|---|---|
| **Q. 3.** | **(a)** Describe the concept of an entity, attribute, and relationship in an ER model. Construct an Entity-Relationship (ER) diagram for a Library Management System. The system should maintain information about books, authors, members, and book issues/returns. | **[05]** | **CO1 L2** |
| | **(b)** Apply the algorithm for disjunctive selection to σPrice<500 ∨ Category='Electronics' (PRODUCT), explaining how the qualifying records can be retrieved and combined. | **[05]** | **CO3 L3** |
| **Q.2.** | **(a)** Differentiate between DDL (Data Definition Language) and DML (Data Manipulation Language) in SQL. Mention and explain at least four SQL commands, giving suitable examples for each command. | **[05]** | **CO2 L2** |
| | **(b)** Differentiate between serial and non-serial schedules, and discuss how serializability ensures correctness during concurrent transaction execution. | **[05]** | **CO3 L2** |
| **Q.3.** | **(a)** Compare 2NF and 3NF and identify the type of redundancy removed by each. | **[05]** | **CO2 L2** |
| | **(b)** Describe the implementation of the timestamp-based protocol in database transaction scheduling with a suitable example. | **[05]** | **CO4 L2** |

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# Yeshwantrao Chavan College of Engineering
## Department of Computer Science and Engineering
### Re-Mid Semester Examination
### B. Tech. 5th Semester Session 2026-2027(ODD)
### Cyber Crime Administration(23OE3528)

**Time: 1 hr. 30 min** **Max Marks: 30**

**Note :-1. All questions are compulsory** **Date: 05/11/2026**

| Que. No. | | Max Marks | CO | Level |
|---|---|---|---|---|
| Q.1 | A) Describe the major stages in the evolution of cybercrime and explain the changes in cyber threats over time. | 5 | CO1 | L2 |
| | B) Differentiate between the roles and responsibilities of MHA, CERT-In, and NCIIPC in preventing and managing cybercrime in India, with a suitable example for each. | 5 | CO4 | L4 |
| Que. No. | | | | |
| Q.2 | A) Explain the different types of digital evidence and describe the procedures followed for collecting, preserving, analyzing, and presenting digital evidence during a cybercrime investigation. | 5 | CO2 | L2 |
| | B ) Analyze the digital forensic investigation lifecycle and its application in cybercrime investigation. | 5 | CO5 | L4 |
| Que. No. | | | | |
| Q.3 | A) Explain the relevant sections of the IPC and CrPC applicable to cyber offences, with one real-life example. | 5 | CO3 | L3 |
| | B) A financial institution faces AI-generated phishing attacks and uses machine learning-based threat detection to identify suspicious activities. Analyze the role of AI/ML in cyberattacks and cybersecurity defense by referring to a suitable real-life Indian or international cybercrime case study. | 5 | CO5 | L5 |

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# Yeshwantrao Chavan College of Engineering
## Department of Computer Science and Engineering
### Mid Semester Examination-I
### B. Tech. 5th Semester Session 2026-2027(ODD)
### Cyber Crime Administration(23OE3528)

**Time: 1 hr. 30 min** **Max Marks: 30**

**Note :-1. All questions are compulsory** **Date: 29/10/2026**

| Que. No. | Solve any Two | Max Marks | CO | Level |
|---|---|---|---|---|
| Q.1 | A) Analyze the roles of MHA, CERT-In, and NCIIPC in the administration and prevention of cybercrime in India. How do these agencies contribute to strengthening the national cybersecurity framework? | 5 | CO4 | L4 |
| | B) Evaluate the key provisions of the National Cyber Security Policy in India and assess how effectively they help organizations prevent and respond to cyber threats. | 5 | CO4 | L5 |
| | C) Categorize the roles of appropriate cybercrime authorities in handling cyberattacks affecting a bank account, a private company's network, and a critical infrastructure system, and describe how each authority can assist in investigating the incidents. | 5 | CO4 | L4 |
| **Que. No.** | **Solve any Two** | | | |
| Q.2 | A) Describe the purpose and typical use of EnCase, FTK, Autopsy, and Wireshark. Also distinguish email, network, mobile, and cloud forensics. | 5 | CO5 | L3 |
| | B) Identify the major challenges in collecting, preserving, and examining digital evidence from cloud environments. Describe how investigators can handle these challenges to maintain the reliability of cloud forensic investigations. | 5 | CO5 | L3 |
| | C) Describe the role of digital forensics in investigating cybercrime and summarize how forensic investigation helps in identifying, examining, and documenting digital evidence. | 5 | CO5 | L3 |
| Que. No. | Solve any Two | | | |
| Q.3 | A) Describe the major current and future cybercrime trends, such as ransomware, deepfakes, cyber warfare, and cryptocurrency-related crimes. Analyse their potential impact on organizations and society. | 5 | CO5 | L4 |
| | B) Analyze the role of AI/ML in modern cyberattacks and cybersecurity defense with reference to a famous Indian or international cybercrime case study. Illustrate how AI/ML techniques were used by attackers or defenders in the case. | 5 | CO2 | L4 |
| | c) A university experiences phishing attempts, AI-generated impersonation content, account compromises, and a ransomware incident. It has no formal cybercrime policy, and students and employees are unaware of reporting and response procedures. As a member of the cybersecurity committee, formulate an institutional cybercrime policy covering prevention, awareness, reporting, incident response, backup, responsibilities, and measures for emerging threats | 5 | CO5 | L5 |

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